

Modelo de la planta:  $P(s) = \frac{k}{(s+p_1)(s+p_2)}$

Discreto:  $P_d(z) = \frac{b}{z^2 + a_1 z + a_2} = \frac{Y(z)}{X(z)}$

$$\rightarrow Y(z)(z^2 + a_1 z + a_2) = b X(z)$$

$$Y(k+2) + a_1 Y(k+1) + a_2 Y(k) = b X(k)$$

$$Y(k+2) = -a_1 Y(k+1) - a_2 Y(k) + b X(k)$$

$$Y(m) + a_1 Y(m-1) + a_2 Y(m-2) = b X(m-2)$$

$$\frac{(18.111+4.4409e-16i)}{s^2 + 14.26s + 53.02}$$

$\rightarrow$  com ARX

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$$(s^2 + 14.26s + 53.02)$$

$\rightarrow$  com CM